

Show understanding that the DBMS carries out all creation/modification of the database structure using its Data Definition Language (DDL)

Show understanding that the DBMS carries out all queries and maintenance of data using its DML

Show understanding that the industry standard for both DDL and DML is Structured Query Language (SQL)

9618_s23_qp_12_sc_2.c.iii

QUESTION

- (iii) The following SQL script should return the number of riders that have the rider level beginner and have a lesson booked on 09/09/2023.

```
SELECT SUM(STUDENT.RiderLevel) AS NumberOfRiders
```

```
FROM STUDENT, LESSON
```

```
WHERE StudentID = StudentID
```

```
OR Date = #09/09/2023#
```

```
AND STUDENT.RiderLevel = Beginner;
```

There are **four** errors in the script.

Identify **and** correct each error.

- 1
- 2
- 3
- 4

[4]

MARK SCHEME

2(c)(iii)	1 mark each - SUM should be COUNT // SELECT COUNT(STUDENT.RiderLevel) - The WHERE statement needs the table names before each field name // WHERE STUDENT.StudentID = LESSON.StudentID - The OR should be AND // AND Date = #09/09/2023# - Beginner is missing the speech marks // STUDENT.RiderLevel = "Beginner";	4
-----------	--	----------

Understand given SQL (DDL) statements and be able to write simple SQL (DDL) statements using a sub-set of statements

9618_w25_qp_13_sc_5.b

QUESTION

(b) Some example data from the `STAFF` table is shown.

StaffID	StaffFirstName	StaffLastName	Department	RemoteWorker
1	Ralph	Jura	E	Yes
2	Luca	Emcee	A	Yes
3	Darwin	Acula	F	No

Write a Structured Query Language (SQL) script to define the table `STAFF`.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

5(b)	1 mark per bullet point, max 4 marks • Create table with opening and closing brackets and commas separating the attributes • Appropriate data types for StaffFirstName, StaffLastName and Department • Appropriate data types for StaffID and RemoteWorker • Primary key correctly defined Example Answer 1: <pre>CREATE TABLE STAFF(StaffID INTEGER, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN, PRIMARY KEY (StaffID));</pre> Example Answer 2: <pre>CREATE TABLE STAFF(StaffID INTEGER NOT NULL PRIMARY KEY, StaffFirstName VARCHAR, StaffLastName VARCHAR, Department CHAR, RemoteWorker BOOLEAN);</pre>	4
------	---	----------

9618_w25_qp_12_sc_4.b

QUESTION

(b) The table CONTAINER needs an additional field to store the data for the last inspection date.

Write a Structured Query Language (SQL) script to add **one** field to the table CONTAINER to store the date of last inspection of the container, for example 08/07/2019.

.....

.....

.....

..... [2]

MARK SCHEME

4(b)	1 mark per bullet point, max 2 marks • ALTER TABLE statement • Including the additional field with a suitable field name and suitable field type Example answer <pre>ALTER TABLE CONTAINER ADD InspectionDate DATE;</pre>	2
------	--	----------

9618_w24_qp_13_sc_4.c.i

QUESTION

(c) The table definition for `BATCH` is repeated here:

BATCH(BatchID, Type, Flavour, Size, SellingPrice, EndDate)

Sample data for the table `BATCH` is given:

BatchID	Type	Flavour	Size	SellingPrice	EndDate
KIV12	Plain	Vanilla	1	2.20	12/12/2024
RIC14	Plain	Chocolate	0.5	1.80	12/12/2024
TYL1	Non-dairy	Lemon	0.5	2.10	01/02/2025
FYV2	Non-dairy	Vanilla	0.25	1.50	02/02/2025
BIV13	Plain	Vanilla	1	2.20	02/02/2024

(i) Write an SQL script to define the table BATCH.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

..... [5]

MARK SCHEME

4(c)(i)	1 mark for each bullet point • Create table <code>BATCH</code> with opening and closing brackets, all statements within brackets • <code>BatchID</code>, <code>Type</code> and <code>Flavour</code> as <code>varchar</code> or equivalent • ...with suitable constraint(s) • <code>Size</code> as <code>decimal</code>, <code>SellingPrice</code> as <code>currency</code>, <code>EndDate</code> as <code>date</code> (or equivalent) • Primary key identified as <code>BatchID</code> e.g. <pre>CREATE TABLE BATCH(BatchID VARCHAR(6) NOT NULL, Type VARCHAR (20) NOT NULL, Flavour VARCHAR (20) NOT NULL, Size FLOAT, SellingPrice CURRENCY, EndDate DATE, PRIMARY KEY (BatchID));</pre>	5
---------	---	----------

QUESTION

- (ii) The table shows sample data for the table `REPAIR_PART`.

PartID	RepairNumber	Quantity
ACD128SA	0022	3
PPOR543DWW	0022	1
TR453	0023	1
PPOR543DWW	0023	2
WED5	0024	5

Write a Structured Query Language (SQL) script to define the table `REPAIR_PART`.

Include constraints (restrictions) on the data that can be entered into each field where appropriate.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

MARK SCHEME

2(b)(ii)	1 mark for each bullet point • Create table <code>REPAIR_PART</code> with opening and closing brackets, all statements inside the brackets • <code>PartID</code> and <code>RepairNumber</code> as <code>varchar</code> (or equivalent) • <code>Quantity</code> as <code>integer</code> • At least 1 appropriate constraint • Dual primary key For example: <pre>CREATE TABLE REPAIR_PART(PartID VARCHAR(20) NOT NULL, RepairNumber VARCHAR(4) NOT NULL, Quantity INT NOT NULL, PRIMARY KEY (PartID, RepairNumber));</pre>	5
----------	--	---

9618_s24_qp_13_sc_4.b

QUESTION

(b) Sample data for the table `PERFORMANCE` is shown:

PerformanceID	ShowID	ShowDate	StartTime
0001	MK12	5/5/2025	13:00
0002	MK12	5/5/2025	19:30
0003	MK12	6/5/2025	19:00
0004	OP3	7/5/2025	18:30
0005	OP3	8/5/2025	18:30
0006	OP3	9/5/2025	13:00

Write a Structured Query Language (SQL) script to define the table `PERFORMANCE`.

[4]

MARK SCHEME

4(b)	1 mark each for: • Creating table <code>PERFORMANCE</code> with opening and closing brackets • Setting all four attributes with appropriate data types • Setting <code>PerformanceID</code> as primary key • Setting <code>ShowID</code> as foreign key referencing <code>SHOW</code> table Example: <pre>CREATE TABLE PERFORMANCE (PerformanceID varchar NOT NULL, ShowID varchar, ShowDate Date, StartTime Time, PRIMARY KEY (PerformanceID), FOREIGN KEY (ShowID) REFERENCES SHOW (ShowID));</pre>	4
------	--	---

9618_s24_qp_12_sc_4.b

QUESTION

(b) Sample data for the table `EXAM` is shown:

ExamID	Subject	Level	TotalMarks
00956124	Computer Science	2	75
00956125	Computer Science	3	120
00956126	Mathematics	2	100
00956127	Mathematics	3	150
00956128	Physics	2	70
00956129	Physics	3	80

Write a Structured Query Language (SQL) script to define the table `EXAM`.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

4(b)	1 mark each: • Creating table EXAM with opening and closing brackets • All fields with appropriate data types and commas at end of lines • ExamID as primary key Example: <pre>CREATE TABLE EXAM(ExamID varchar NOT NULL, Subject varchar, Level int, TotalMarks int, PRIMARY KEY (ExamID));</pre>	3
------	---	----------

9618_s24_qp_12_sc_4.c

QUESTION

(c) The table EXAM_QUESTION has been created but the foreign key has not been linked.

Write an SQL script to update EXAM_QUESTION and link the foreign key to EXAM.

.....

.....

.....

..... [2]

MARK SCHEME

4(c)	1 mark each: • Altering table EXAM_QUESTION • Linking ExamID to ExamID in EXAM Example. <pre>ALTER TABLE EXAM_QUESTION ADD FOREIGN KEY (ExamID) REFERENCES EXAM (ExamID);</pre>	2
------	--	----------

QUESTION

- (c) The company has another database, `FARMING`, for a different game.

The database `FARMING` has a table named `EVENT` which is shown with some sample data.

PlayerID	EventID	Category	Points
000123	3	Build	100
000124	1	Grow	36
000123	4	Grow	22
000123	7	Create	158
000125	3	Grow	85
000125	4	Build	69

- (i) The database `FARMING` has a second table created named `PLAYER` that has the primary key `PlayerID`.

The field `PlayerID` in `EVENT` needs to be set up as a foreign key to link to `PlayerID` in `PLAYER`.

Write a Structured Query Language (SQL) script to change the table definition for `EVENT` to link the foreign key to `PLAYER`.

.....

 [2]

MARK SCHEME

6(c)(i)	1 mark for each correct clause: - Alter table <code>EVENT</code> - Adding foreign key as <code>PlayerID</code> referencing correct table <pre>ALTER TABLE EVENT ADD FOREIGN KEY (PlayerID) REFERENCES PLAYER (PlayerID);</pre>	2
---------	--	----------

QUESTION

- (c) (i) Write a Structured Query Language (SQL) script to define the database called `SHOP`.

.....
 [1]

MARK SCHEME

3(c)(i)	1 mark for <code>CREATE DATABASE SHOP;</code>	1
---------	---	----------

9618_s23_qp_11_sc_2.b.iii

QUESTION

(iii) Part of the database table `BIRD_TYPE` is shown:

BirdID	Name	Size
0123	Blackbird	Medium
0035	Jay	Large
0004	Raven	Large
0085	Robin	Small

The database only supports these data types:

- character
- varchar
- Boolean
- integer
- real
- date
- time

Write a Structured Query Language (SQL) script to define the table `Bird_Type`.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

2(b)(iii)	1 mark each • CREATE TABLE start and end bracket • Bird ID as CHAR/VARCHAR • Name and size as VARCHAR/CHAR • Bird ID as primary key Example answer: <pre>CREATE TABLE BIRD_TYPE (BirdID CHAR(4) NOT NULL, Name VARCHAR(9), Size VARCHAR(6), PRIMARY KEY (BirdID));</pre>	4
-----------	---	----------

9618_w22_qp_13_sc_2.d

QUESTION

(d) Write the SQL script to add **one** field to the table PHOTOGRAPH to store the resolution of the photograph, e.g. 1920 × 1068.

.....

.....

.....

..... [2]

MARK SCHEME

2(d)	1 mark for each bullet point: • ALTER TABLE PHOTOGRAPH • ADD Resolution TEXT; <pre>ALTER TABLE PHOTOGRAPH ADD Resolution TEXT / VARCHAR(11);</pre>	2
------	---	----------

QUESTION

- (d) Write an SQL script to include two new fields in `CAMERA_DATA` to store the number of photographs currently on the camera **and** the date the camera was last used.

.....
.....
.....
.....
.....
..... [3]

MARK SCHEME

4(d)	1 mark for each bullet point: • <code>ALTER TABLE CAMERA_DATA</code> • <code>ADD NumberStored INTEGER</code> • <code>, LastUsed DATE;</code> <code>ALTER TABLE CAMERA_DATA</code> <code>ADD NumberStored INTEGER, LastUsed DATE;</code>	3
------	--	----------

QUESTION

- (ii) Write DDL statements to include a field in the table `PURCHASE` to store the date of the order.

.....
.....
.....
..... [3]

MARK SCHEME

6(c)(ii)	1 mark per bullet point - ALTER TABLE PURCHASE - ADD OrderDate - Suitable data type, e.g. DATE <pre>ALTER TABLE PURCHASE ADD OrderDate DATE;</pre>	3
----------	---	----------

9618_s21_qp_12_sc_1.c.i

QUESTION

(c) Example data from the table RENTAL are given:

RentalID	CustomerID	HouseID	MonthlyCost	DepositPaid
1	22	15B5L	1000.00	Yes
2	13	3F	687.00	No
3	1	12AB	550.00	Yes
4	3	37	444.50	Yes

(i) Complete the following Data Definition Language (DDL) statement to define the table RENTAL.

```
CREATE ..... (
    RentalID INTEGER NOT NULL,
    CustomerID INTEGER NOT NULL,
    HouseID ..... (5) NOT NULL,
    MonthlyCost ..... NOT NULL,
    DepositPaid BOOLEAN NOT NULL,
    ..... (RentalID)
);
```

[4]

MARK SCHEME

1(c)(i)	1 mark for each correctly completed line <pre>CREATE TABLE RENTAL(RentalID INTEGER NOT NULL, CustomerID INTEGER NOT NULL, HouseID VARCHAR (5) NOT NULL, MonthlyCost REAL/CURRENCY NOT NULL, DepositPaid BOOLEAN NOT NULL, PRIMARY KEY (RentalID));</pre>	4
---------	---	----------

9618_s21_qp_11_sc_7.b.iii

QUESTION

(iii) Bobby wants to name his database SHOPORDERS.

Write a Data Definition Language (DDL) statement to define a new database with the name SHOPORDERS.

.....

..... [1]

MARK SCHEME

7(b)(iii)	1 mark <pre>CREATE DATABASE SHOPORDERS;</pre>	1
-----------	---	----------

Write an SQL script to query or modify data (DML) which are stored in (at most two) database tables

9618_w25_qp_13_sc_5.c

QUESTION

(c) Products are given a rating between 1 and 10 inclusive.

Write an SQL script to return only the `ProductID`, `ProductName` and `ComplaintDetails` for all products with a rating of 5 or less. The results need to be displayed in descending order of rating.

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

MARK SCHEME

5(c)	1 mark per bullet point, max 5 marks • SELECT and the correct attributes • FROM and the correct tables • Tables joined correctly • Correct condition for the rating • Correct ORDER BY clause Example Answer 1 SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT, COMPLAINT WHERE PRODUCT.ProductID = COMPLAINT.ProductID AND Rating <=5 ORDER BY Rating DESC; Example Answer 2 SELECT PRODUCT.ProductID, ProductName, ComplaintDetails FROM PRODUCT INNER JOIN COMPLAINT ON PRODUCT.ProductID = COMPLAINT.ProductID WHERE Rating <=5 ORDER BY Rating DESC;	5
------	---	----------

9618_w25_qp_12_sc_4.c

QUESTION

- (c) Write an SQL script to return the number of containers stored in the database for the ship with the name Caledonia.

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

4(c)	1 mark per bullet point, max 4 marks • SELECT <u>COUNT</u> statement • Using the correct tables • Joining the tables • The condition for the name Example Answer 1: <pre>SELECT COUNT(ContainerID) FROM CONTAINER, SHIP WHERE CONTAINER.ShipID = SHIP.ShipID AND ShipName = "Caledonia";</pre> Example Answer 2: <pre>SELECT COUNT(ContainerID) FROM CONTAINER INNER JOIN SHIP ON CONTAINER.ShipID = SHIP.ShipID WHERE ShipName = "Caledonia";</pre>	4
------	---	---

9618_w25_qp_11_sc_2.c.i

QUESTION

(c) Some example data from the `PLACEMENT` table is shown:

PlacementID	StudentID	CompanyID	StartDate	EndDate	Complete
PC001	LDEA01	MSCM	01/01/2023	12/07/2023	TRUE
PC002	LDEA01	MSCM	01/04/2024	08/05/2024	FALSE
PC003	ALAU02	NEAM	09/03/2020	10/03/2021	FALSE
PC004	LOLI75	GOUZ	07/06/2018	11/09/2018	TRUE

- (i) Write a Structured Query Language (SQL) script to delete all placements that have been completed.

.....

.....

.....

..... [2]

MARK SCHEME

2(c)(i)	1 mark per bullet point, max 2 marks - DELETE FROM and correct table - Correct condition Example answer: DELETE FROM PLACEMENT WHERE Complete = TRUE;	2
---------	--	----------

9618_w25_qp_11_sc_2.c.ii

QUESTION

- (ii) Write an SQL script to return the total number of placements completed by the student with ID LDEA01 at the company with ID NEAM. The total should be given an appropriate name.

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

2(c)(ii)	1 mark per bullet point, max 4 marks - SELECT COUNT any appropriate field - AS and an appropriate name - FROM and correct table and WHERE and one correct condition - Two AND clauses with the other two correct conditions Example answer: SELECT COUNT (CompanyID) AS TotalPlacements FROM PLACEMENT WHERE CompanyID = "NEAM" AND StudentID = "LDEA01" AND Complete = TRUE;	4
----------	--	----------

QUESTION

- (c) The attribute `Collected` in the table `ORDER` stores the Boolean value `TRUE` if the order has been collected and `FALSE` if the order has not been collected.

Write an SQL script to return the customer name for each customer that has orders they have not collected. Include the number of orders each customer has not collected with an appropriate title.

An example output might be:

CustomerName	NotCollected
Jack Wright	2
Lin Cho	1
Santaya Yui	1

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

6(c)	1 mark each • Selection of customer name and counting any field from ORDER as an appropriate identifier • Joining tables ORDER and CUSTOMER • AND (or WHERE) clause: Collected = FALSE • Grouping by customer ID or customer name Example 1: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER, CUSTOMER WHERE ORDER.CustomerID = CUSTOMER.CustomerID AND Collected = FALSE GROUP BY CUSTOMER.CustomerID Example 2: SELECT CustomerName, COUNT(OrderID) AS NotCollected FROM ORDER INNER JOIN CUSTOMER ON ORDER.CustomerID = CUSTOMER.CustomerID WHERE Collected = FALSE GROUP BY CUSTOMER.CustomerID	4
------	---	----------

9618_s25_qp_12_sc_5.d.i

QUESTION

- (d) (i) Write a Structured Query Language (SQL) script to count the number of items purchased by the user with the username "KAT123".

.....

.....

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

5(d)(i)	1 mark for each point • Select COUNT • FROM (using correct tables) and joining • WHERE (or AND) condition for correct username Example 1: SELECT COUNT(ItemName) FROM CHARACTER_ITEM INNER JOIN CHARACTER ON CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID WHERE CHARACTER.Username = "KAT123"; Example 2: SELECT COUNT(ItemName) FROM CHARACTER_ITEM, CHARACTER WHERE CHARACTER.CharacterID = CHARACTER_ITEM.CharacterID AND CHARACTER.Username = "KAT123";	3
---------	---	----------

9618_s25_qp_12_sc_5.d.ii

QUESTION

(ii) The following changes need to be made to the character with the ID "0002":

- level changed to 3
- money changed to 10000.00

Write an SQL script to change the character's data.

.....

.....

.....

.....

.....

.....

.....

[3]

MARK SCHEME

5(d)(ii)	1 mark for each point • UPDATE character • SET level and money • WHERE condition e.g. UPDATE CHARACTER SET Level = 3, Money = 10000.00 WHERE CharacterID = "0002";	3
----------	--	----------

9618_s25_qp_11_sc_5.e

QUESTION

- (e) Write an Structured Query Language (SQL) script to output the customer ID, the customer's name and the total cost of the customer's orders that have **not** been paid.

The output of the total cost must have an appropriate title.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

5(e)	1 mark each • Selecting the customer ID, customer name and sum of TotalCost with appropriate identifier • FROM clause with suitable join of tables (ON or WHERE) • ORDER.Paid = FALSE condition (with correct key word) • GROUP BY condition e.g. SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER INNER JOIN ORDER ON CUSTOMER.CustomerID = ORDER.CustomerID WHERE ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID; Alternative JOIN Statement SELECT CUSTOMER.CustomerID, CUSTOMER.Name, Sum(ORDER.TotalCost) AS TotalOwed FROM CUSTOMER,ORDER WHERE CUSTOMER.CustomerID = ORDER.CustomerID AND ORDER.Paid = FALSE GROUP BY CUSTOMER.CustomerID;	4
------	---	----------

9618_w24_qp_13_sc_4.b

QUESTION

- (b) Write an SQL script to return the total quantity of ice cream sold to the customer with the ID of 0034E in the year 2023.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

4(b)	1 mark each: • <code>SELECT SUM(Quantity)</code> • <code>FROM SALE WHERE</code> and one correct condition • <code>AND</code> with remainder correct conditions e.g. <pre>SELECT SUM(Quantity) FROM SALE WHERE CustomerID = "0034E" AND Date >= #01/01/2023# AND Date <= #31/12/2023#;</pre>	3
------	--	----------

9618_w24_qp_12_sc_6.b.ii

QUESTION

- (ii) The database also has the table `INVOICE` that stores data about each invoice that is sent to a customer.

Example data from the table `INVOICE` is given.

InvoiceID	DateSent	Amount	Paid	JobID
29262	12/12/2023	105.20	Y	221
26765	11/11/2023	200.00	Y	315
13290	02/01/2024	50.00	Y	315
34090	05/02/2024	25.95	N	569

Write a Structured Query Language (SQL) script to return the total amount of all the invoices sent in the year 2023 that have been paid.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

6(b)(ii)	1 mark each • Select SUM of Amount • From the correct table and one correct condition • Remaining correct condition Example: <pre>SELECT SUM(Amount) FROM INVOICE WHERE Paid = "Y" AND DateSent >= #01/01/2023# AND DateSent <= #31/12/2023#</pre>	3
----------	--	----------

9618_w24_qp_11_sc_2.b.iii

QUESTION

- (iii) Suppliers send invoices to the company for the parts that are used. A new table, `INVOICE`, stores the data about each invoice and whether it has been paid or not.

The design for the table `INVOICE` is shown:

```
INVOICE(InvoiceID, SupplierID, AmountDue, Paid, DatePaid)
```

The table shows sample data for the table `INVOICE`.

InvoiceID	SupplierID	AmountDue	Paid	DatePaid
000001	JK675	22.50	TRUE	01/01/2024
000002	WR443	358.99	FALSE	
000003	JK675	10.21	FALSE	

Write an SQL script to return the total amount due to the supplier with the ID of JK675 for all the invoices that have **not** currently been paid.

.....

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

2(b)(iii)	1 mark for each bullet point • Select SUM of AmountDue • From the correct table and one correct condition • Second correct condition <pre>SELECT SUM(AmountDue) FROM INVOICE WHERE SupplierID = "JK675" AND Paid = FALSE;</pre>	3
-----------	--	----------

9618_s24_qp_13_sc_4.c

QUESTION

9

- (c) Write an SQL script to return the number of times each show is scheduled. For example, in the sample data in part (b), the show MK12 is scheduled three times.

The result needs to include the show name and a suitable field name for the number of times it is scheduled.

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

4(c)	1 mark each for: • Selecting COUNT of an attribute in PERFORMANCE table with suitable name • FROM clause • Joining tables • Grouping by the title and selecting the title Example 1: SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE, SHOW WHERE PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title; Example 2: SELECT SHOW.Title, Count (PERFORMANCE.PerformanceID) AS NumberOfShowings FROM PERFORMANCE INNER JOIN SHOW ON PERFORMANCE.ShowID = SHOW.ShowID GROUP BY SHOW.Title;	4
------	---	---

9618_s24_qp_11_sc_6.c.ii

QUESTION

(ii) Write an SQL script to return the number of events that each player has completed.

.....

.....

.....

.....

.....

..... [3]

MARK SCHEME

6(c)(ii)	1 mark each: • Selecting PlayerID from EVENT • Counting EventID • Grouping by the PlayerID Example: SELECT PlayerID, COUNT (EventID) FROM EVENT GROUP BY PlayerID;	3
----------	---	---

QUESTION

- (ii) Write the Structured Query Language (SQL) script to return the total number of courses that have started after 9 September 2023.

The value returned must have an appropriate field name.

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

3(b)(ii)	1 mark for each bullet point. • <code>SELECT Count (CourseID)</code> • <code>AS NumOfCourses</code> • <code>FROM COURSE_SCHEDULE</code> • <code>WHERE DateStarted > "09/09/23";</code> <code>SELECT Count (CourseID) AS NumOfCourses FROM</code> <code>COURSE_SCHEDULE</code> <code>WHERE DateStarted > "09/09/23";</code>	4
----------	--	----------

QUESTION

- (ii) Write the SQL script to return the total quantity of items that the customer with the ID of HJ231 has ordered.

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

3(c)(ii)	1 mark for each line (max 4) Either: <pre>SELECT SUM(Quantity) FROM ORDER_ITEM, SHOP_ORDER WHERE ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo AND SHOP_ORDER.CustomerID = 'HJ231';</pre> OR <pre>SELECT SUM(Quantity) FROM ORDER_ITEM (INNER) JOIN SHOP_ORDER ON ORDER_ITEM.OrderNo = SHOP_ORDER.OrderNo WHERE SHOP_ORDER.CustomerID = 'HJ231';</pre>	4
----------	---	----------

9618_s23_qp_12_sc_2.c.ii

QUESTION

- (ii) Write a Structured Query Language (SQL) script to return the names of all the horses that have the horse level intermediate or beginner.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

MARK SCHEME

2(c)(ii)	1 mark each • SELECT field Name • FROM table HORSE • WHERE with Intermediate / Beginner • OR with Beginner / Intermediate Example answer: <pre>SELECT Name FROM HORSE WHERE HorseLevel = "Intermediate" OR HorseLevel = "Beginner";</pre>	4
----------	---	----------

QUESTION

- (iii) The following SQL script should return the number of riders that have the rider level beginner and have a lesson booked on 09/09/2023.

```
SELECT SUM(STUDENT.RiderLevel) AS NumberOfRiders
FROM STUDENT, LESSON
WHERE StudentID = StudentID
OR Date = #09/09/2023#
AND STUDENT.RiderLevel = Beginner;
```

There are **four** errors in the script.

Identify **and** correct each error.

- 1
- 2
- 3
- 4

[4]

MARK SCHEME

2(c)(iii)	1 mark each - SUM should be COUNT // SELECT COUNT(STUDENT.RiderLevel) - The WHERE statement needs the table names before each field name // WHERE STUDENT.StudentID = LESSON.StudentID - The OR should be AND // AND Date = #09/09/2023# - Beginner is missing the speech marks // STUDENT.RiderLevel = "Beginner";	4
-----------	---	----------

QUESTION

(iv) The database tables are repeated here for reference:

BIRD_TYPE(BirdID, Name, Size)

BIRD_SEEN(SeenID, BirdID, Date, Location, PersonID)

PERSON(PersonID, FirstName, LastName, EmailAddress)

Complete the SQL script to return the number of birds of each size seen by the person with the ID of J_123.

```
SELECT BIRD_TYPE.Size, ..... (BIRD_TYPE.BirdID)

      AS NumberOfBirds

FROM BIRD_TYPE, .....

WHERE ..... = "J_123"

AND BIRD_TYPE.BirdID = .....

..... BIRD_TYPE.Size;
```

[5]

MARK SCHEME

2(b)(iv)	1 mark for each correctly completed space <pre>SELECT BIRD_TYPE.Size, COUNT(BIRD_TYPE.BirdID) AS NumberOfBirds FROM BIRD_TYPE, BIRD_SEEN WHERE BIRD_SEEN.PersonID = "J_123" AND BIRD_TYPE.BirdID = BIRD_SEEN.BirdID GROUP BY BIRD_TYPE.Size;</pre>	5
----------	--	----------

QUESTION

(c) Complete the SQL script to return the total number of telescopes owned by the company whose ID begins with HW.

```
SELECT ..... (.....)

FROM TELESCOPE

WHERE ..... LIKE .....
```

[4]

MARK SCHEME

2(c)	1 mark for each correctly completed missing part: SELECT <u>COUNT</u> (<u>TelescopeID</u>) FROM TELESCOPE WHERE <u>CompanyID</u> LIKE 'HW%';	4
------	---	----------

9618_w22_qp_12_sc_5.b

QUESTION

- (b) Write the Structured Query Language (SQL) script to add a new record in the table **TREE** to store the following data:

Attribute	Value
TreeID	LOW_1276
ScientificName	Salix_Alba
MaxHeight	30.00
FastGrowing	TRUE

.....

 [3]

MARK SCHEME

5(b)	1 mark for each bullet point: • INSERT INTO TREE • VALUES () and correct values • Values in correct order Option 1: INSERT INTO TREE(TreeID, ScientificName, MaxHeight, FastGrowing) VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE); Option 2: INSERT INTO TREE VALUES('LOW_1276', 'Salix_Alba', 30.00, TRUE);	3
------	--	----------

9618_w22_qp_11_sc_4.c.ii

QUESTION

- (ii) Complete the Structured Query Language (SQL) script to display the total number of photographs that have been taken using a camera with a camera ID starting with CAN.

```
SELECT ..... (.....)

FROM .....

WHERE CameraID LIKE ..... ;
```

[4]

MARK SCHEME

4(c)(ii)	1 mark for each correctly completed empty space: • COUNT • PhotoID • PHOTO_DATA • 'CAN*' // 'CAN%' <pre>SELECT COUNT(PhotoID) FROM PHOTO_DATA WHERE CameraID LIKE 'CAN*'; // WHERE CameraID LIKE 'CAN%';</pre>	4
----------	--	----------

9618_w21_qp_12_sc_6.c.i

QUESTION

- (c) The shop manager uses both Data Definition Language (DDL) and Data Manipulation Language (DML) statements to create and search the database.

- (i) Complete the DML statements to return the total number of items purchased with the purchase ID of 3011A.

```
SELECT SUM(.....)

FROM .....

WHERE ..... = ..... ;
```

[4]

MARK SCHEME

6(c)(i)	1 mark for each correctly completed space <pre>SELECT SUM(Quantity) FROM PURCHASE_ITEM WHERE PurchaseID = "3011A";</pre>	4
---------	--	----------

9618_w21_qp_11_sc_5.c.i

QUESTION

(c) Javier uses Data Definition Language (DDL) and Data Manipulation Language (DML) statements in his database.

(i) Complete the following DML statements to return the number of cars for sale in each shop.

SELECT COUNT (.....)

FROM

..... ShopID

[3]

MARK SCHEME

5(c)(i)	1 mark per correctly completed statement SELECT COUNT (RegistrationNumber) FROM CAR GROUP BY ShopID	3
---------	---	----------

9618_w21_qp_11_sc_5.c.ii

QUESTION

(ii) Complete the DML statement to include the following car in the table CAR.

Field	Data
RegistrationNumber	123AA
Make	Tiger
Model	Lioness
NumberOfMiles	10500
ShopID	12BSTREET

..... CAR

..... ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")

[2]

MARK SCHEME

5(c)(ii)	1 mark for each correct statement INSERT INTO CAR VALUES ("123AA", "Tiger", "Lioness", 10500, "12BSTREET")	2
----------	---	----------

QUESTION

- (ii) Write a Data Manipulation Language (DML) script to return the first name and last name of all customers who have **not** paid their deposit.

.....
.....
.....
.....
.....
..... [4]

MARK SCHEME

1(c)(ii)	1 mark per bullet point • Select FirstName and LastName • From both tables • Where DepositPaid = No • Joining tables (either AND, or INNER JOIN) Example script: SELECT FirstName, LastName FROM CUSTOMER, RENTAL WHERE DepositPaid = No AND RENTAL.CustomerID = CUSTOMER.CustomerID;	4
----------	---	----------